SEMPY01 - FINAL ASSESSMENT

Submission Requirements

- Submit a **single PDF report** (exported from Markdown/Word/LaTeX).
- Submit your **code** as a `.zip` archive.
- All plots must be generated computationally.
- Axes must include **units**.
- Each section starts on a **new page**.
- Every result must include a **Reflection / Discrepancy Analysis** subsection.
- Final deliverable includes a **10–15 minute presentation**.

Task 1 — Drift and Diffusion Currents

Implement the current densities

$$J_{\text{drift}} = qn\mu E, \qquad J_{\text{diff}} = qD\frac{dn}{dx}.$$

Requirements:

1. Plot J_{drift} vs. E for **electrons** and **holes**.
2. Assume a linear carrier gradient and plot J_{diff}.
3. Verify the Einstein relation:

$$\frac{D}{\mu} = \frac{k_B T}{q}.$$

Task 2 — Mobility and Temperature Dependence

Use the model

$$\mu(T) \propto T^{-m},$$

where m reflects the dominant scattering mechanism.

Requirements:

1. Compute and plot $\mu(T)$ for $200\,\text{K} \leq T \leq 600\,\text{K}$.
2. Compare electron vs. hole mobility.

3. Discuss phonon vs. impurity scattering.

Task 3 — Nonequilibrium Excess Carriers

Solve the continuity equation

$$\frac{\partial \Delta n}{\partial t} = G - \frac{\Delta n}{\tau}.$$

Requirements:

1. Plot $\Delta n(t)$ for at least two lifetimes (short vs. long).
2. Compare with literature lifetimes for your semiconductor.

Task 4 — Quasi-Fermi Levels

For a forward-biased p–n junction, compute and plot the quasi-Fermi levels.

Requirements:

1. Plot E_{Fn} and E_{Fp} relative to E_c and E_v.
2. Explain the meaning of the separation:

$$\Delta E_F = E_{Fn} - E_{Fp} = qV_a.$$

Task 5 — Experimental Comparison

Find one real dataset for your semiconductor (mobility μ, diffusion constant D, or lifetime τ).

Requirements:

1. Overlay the experimental dataset with your computed results.
2. Identify one **agreement** and one **discrepancy**.
3. Provide at least one **physical or computational** explanation.

Task 6 — PN Junction Diode

Implement the diode equation

$$I(V) = I_s \left(e^{qV/k_B T} - 1 \right).$$

Requirements:

1. Plot I–V for forward and reverse bias.
2. Compare with a **baseline silicon diode**.
3. Comment on effects of band gap, intrinsic concentration, and carrier properties.

Report Structure (PDF)

1. Introduction

- Identify your semiconductor.
- Explain why transport, mobility, recombination, and quasi-Fermi levels matter.
- Outline Tasks 1–6.

2. Methods

- Present equations used.
- State assumptions (uniform doping, constant effective mass, low-level injection, etc.).
- Include short code snippets where relevant.

3. Results

Include:

- Drift current plot
- Diffusion current plot
- Einstein relation check
- $\mu(T)$ plot
- $\Delta n(t)$ for two lifetimes
- Quasi-Fermi level plot
- Experimental overlay
- Diode I–V curve

4. Discussion

- Summarize insights about mobility, lifetime, transport, quasi-Fermi levels, diode behavior.
- Compare with literature.
- State limitations.

5. Conclusion

- Restate key findings.
- State one thing learned from coding.
- Pose one open question about your semiconductor.

6. References

- IEEE style.
- At least two references (textbook + journal).
- Include PDFs of all references in your submission archive.

Presentation Format (10–15 minutes)

1. **Introduction** — Semiconductor + motivation
2. **Methods** — Equations, assumptions, code structure (A FLOW CHART OF COMPUTATION IS FINE. no need to do a explanation of methods)
3. **Results** — Drift/diffusion, mobility, lifetime, quasi-Fermi levels, diode I–V
4. **Experimental Comparison** — Overlay + commentary
5. **Reflection** — Agreement, discrepancy, limitation
6. **Conclusion** — One next step